CAN Communication Protocol - Electric

1 Introduction

1.1 Purpose

This document specifies the application layer interface protocols and instructions for various types of vehicles based on two-wheel CAN communication. This document aims to improve the safety of various vehicles in operation based on two-wheel CAN communication.

1.2 Document Conventions

This agreement includes PC, central controller, instrument, motor controller, various electronic locks, sensors, batteries and other equipment. This protocol provides the main framework, and the specific data to be uploaded, error types, and other data are provided by each device vendor. This document assumes that all CAN devices understand the basic communication rules of CAN and will not describe the CAN principles.

1.3 Abbreviations

Abbreviations	Meaning
CCU	Central controller, which refers to network communication equipment with GPRS, etc.
ECU	Motor Controller
BLE	Low-power Bluetooth, generally refers to modules with Bluetooth 4.0
SID	Source ID, CAN ID of the sent data
DID	Target ID, CAN ID of received data
CMD	instruction
INDEX	instruct
DLC	Dedicated to indicate CAN data length
DEVID	Device ID, a unique number for each device, 4 bytes in hexadecimal
DEVKEY	Device key, the root key of each device, 16 bytes in hexadecimal
KM	Metric speed unit, kilometers per hour
MPH	English unit of speed, miles per hour
mA	Current unit, milliampere
mV	Voltage unit, millivolt
KM	Metric mileage unit, kilometers
ImA	Battery capacity unit, mAh
Null	Empty, meaning no or 0-filled

2 Protocol Basics

This protocol uses CAN extended frames for communication, mainly describing the contents of the arbitration segment and data segment.

2.1 Bus parameters

Parameter	Value
CAN Type:	29bit extended frame
Baud rate:	250Kbit/S
Maximum Node:	31
Data length:	0-8 bytes
Terminal resistance:	120 ohms

2.2 Arbitration Section

The arbitration segment of the extended frame has 29 bits, which can be divided into 4 bytes, all in hexadecimal.

bit	**bit28-bit24**	**bit23-bit21**	**bit20-bit16**	**bit15-bit8**	**bit7-bit0**
Logo	SID	Null	DID	CMD	Index
definition	Source ID	0 Padding	Target ID	instruction	instruct

2.2.1 Extended Frame

bit	**Logo**	**definition**
bit28 – bit 24	SID	Source ID
bit23 – bit 21	Null	Padding
bit20 – bit 16	DID	Target ID
bit15 – bit 8	CMD	instruction
bit7 – bit0	Index	instruct

2.2.2 ID Allocation

Due to the limitation of the number of bits, the ID can only be in the range of 0x00 to 0x1F, and the smaller the ID number, the higher the priority.

Equipment	**ID**
Main Station	0x00
Battery 1	0x14
Battery 2	0x1D
CCU	0x05
meter	0X0A
ECU	0X12
Charger	0X13
Centralized Controller	0X18
Alarm	0X1E
Broadcast	0X1F

2.2.3 Instruction List

The CMD and Index instructions are used to distinguish the data content, mainly including general instructions, configuration instructions and special instructions. For details, see Section 5. This protocol communication includes two types: point-to-point and broadcast. When DID is 0x00 - 0x1E, it is point-to-point communication, and only DID devices can receive and process it. When DID is 0x1F, it is broadcast communication, and all devices can receive and process it.

2.3 Data segment

The data segment length is 0-8 bytes, consisting of hexadecimal and characters. Data exceeding 8 bytes needs to be sub-packetized. Multi-byte data uses little-endian mode, that is, the low byte comes first and the high byte comes last.

3 Connection rules

All devices communicate directly after power-on.

4 Firmware Upgrade

This document specifies that the CAN communication firmware upgrade uses the Y-modem protocol. The device should be compatible with 128 and 1024 bytes to meet the needs of different devices. The vendor decides whether to perform firmware upgrades in the bootloader or the application, and whether to use a larger flash to back up the old firmware. However, if an upgrade fails, the device must be able to be re-upgraded. If a dual-firmware device fails to upgrade, you can use the old firmware to run it again. The firmware upgrade process remains unchanged. If a single-firmware device fails to upgrade, it will generally stay in the bootloader and send the fault code 0x0A after power-on until it receives a firmware upgrade instruction. All Bootloader communications use plain text. This protocol requires that the firmware file used for upgrading must insert 32 bytes of data before the original firmware bin file, of which the device only uses the last 16 bytes (FWKEY) and defines it by itself, and ignores the first 16 bytes of data during upgrading. The reserved bytes are mainly used by the device to automatically determine whether it can be upgraded. We provide a firmware upgrade packaging tool. Different devices should have different FWKEYs. This FWKEY should be submitted to the provider together with the device, and the provider will manage the FWKEY uniformly. For the detailed process of CAN communication firmware upgrade, please refer to the document "Two-wheel CAN Communication Firmware Upgrade Process". The simplified process is as follows:

1. Devices that require firmware upgrades when powered on.
2. Connect the device to the ready state.
3. Send firmware upgrade command 1A 00.
4. Start firmware upgrade (Y-modem).

   Note: 32 bytes are reserved and placed after the "file length" of the first Y-modem packet.

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5 Contents of the Agreement

The CAN protocol specified in this document mainly includes general instructions, configuration instructions and special instructions. General instructions refer to instructions that can be used by almost all devices, configuration instructions are mainly used to set and read parameters, and special instructions are instructions unique to each device. This protocol stipulates that multi-byte data uses little-endian mode, that is, the low byte is in front and the high byte is in the back. Unless otherwise specified, all data uses unsigned integers. This protocol requires that broadcast data be released periodically until the specified requirements are met. Unless otherwise specified, broadcast data does not require a response. This protocol requires that point-to-point data must be responded to. It only needs to be sent once. If no response is received within the timeout period or the response is abnormal, the sender can define the retransmission mechanism by itself. Except for special instructions, general responses can be used.

5.1 General instructions

SID	DID	CMD	Index	DLC	Data	illustrate
0x00 - 0x1E	0x1F	0x00	0x01	4	D0-D3: Heartbeats	1. Heartbeat starts sending only after it is ready, and stops after it is shut down. 2. Maintain system connection to determine whether the device is online (not unique). Data with CMD 0x00 uses plain text.

SID	DID	CMD	Index	DLC	Data	illustrate
0x00 - 0x1E	0x00 - 0x1E	0x04	CMD	7	D0: CMD initiator D1: Index Initiating D2-D3: Reserved D4-D7: Detailed segment Index D5D6: CRC16 (See Appendix 6.5)	Long packet sending. Initial packet: If a long packet is initiated actively, CMD and Index represent the instructions in the protocol; If it is a response, CMD and Index indicate the received command; Each packet requires a response from the receiver, using a general response. If no response is received within 100ms, the next packet will be sent. NUM represents the current packet number, starting from 1 up to the total number of packets.
0x00 - 0x1F	0x00 - 0x1E	0x0A	R	2-8	D0: Fault code D1-D3: Reserved D4-D7: Detailed exception code	General Response. 1. Except for special requirements, generally only point-to-point instructions are responded to. RES is the basic type of response result. 0x00 is normal, and the others are abnormal. See Appendix 6.3. 2. Detailed exception codes are reserved for device vendors to use.
0x00 - 0x1E	0x1F	0x0E	ERR	1-8	D0: Fault code D1-D3: Reserved D4-D7: Detailed fault code	Fault (plain text). 1. Any device failure can be notified to other devices through this message. ERR is the failure level, see Appendix 6.4. 2. For general fault codes, see Appendix 6.5. 3. For the dedicated fault code, please refer to the equipment protocol. This code is mainly defined and provided by the equipment manufacturer. 4. Detailed fault codes are reserved for equipment suppliers to use at their own discretion.
0x00 - 0x1E	0x00 - 0x1E	0x11	0x01	5	D0-D4: "HWVER"	Read hardware version. 1. The version number is defined by each device manufacturer. It is recommended to use ASCII characters and the length should not exceed 32 characters. 2. Regardless of the length, the response is sent using a long packet.

SID	DID	CMD	Index	DLC	Data	illustrate
0x00 - 0x1E	0x00 - 0x1E	0x11	0x02	5	D0-D4: "FWVER"	Read firmware version. 1. The version number is defined by each device manufacturer. It is recommended to use ASCII characters and the length should not exceed 32 characters. 2. Regardless of the length, the response is sent using a long packet.
0x00 - 0x1E	0x00 - 0x1E	0x11	0x03	2	D0-D1: "SN"	Read Serial Number. 1. The serial number content is specified by a separate file. 2. Regardless of the length, the response is sent using a long packet.
0x00 - 0x0F	0x01 - 0x1E	0x1A	0x00	6	D0-D5: "UPDATE"	Send Firmware Upgrade. 1. Send point-to-point to devices that need firmware updates. 2. The cycle is 100ms, and it is sent 50 times or a response is received from the updated device. 3. Device uses universal answer.
0x00 - 0x0F	0x01 - 0x1E	0x1A	0x01	1-8	Y-modem data packages	Send firmware package (plain text). 1. Specifies that device firmware upgrades use the Y modem protocol. 2. The device should be compatible with 128 and 1024 bytes to meet the needs of different devices. 3. Y-modem data packets are filled in the data segment. 4. The first 32 bytes of the firmware package are inserted into the first Y-modem package "file length". 5. The upgrade file starts from the 33rd byte, and the firmware package is transmitted in plain text. 6. After the device upgrade is complete, it will restart and the host needs to restart the system.
0x01 - 0x1E	0x00 - 0x0F	0x1A	0x02	1-8	Y-modem response	Response firmware package (plain text). 1. The data segment content is the device Y modem response.

SID	DID	CMD	Index	DLC	Data	illustrate
0x00 - 0x0F	0x1F	0x1F	0x1F	7	D0-D6: "SILENCE"	Silence. 1. Instructs all devices to enter silent mode and stop sending broadcast data. This function is mainly used to transmit large data, such as upgrading slave device firmware, downloading logs. 2. In silent mode, the host will not broadcast data such as "RUN". 3. Other devices should not list this situation as a run timeout or heartbeat timeout. 4. In this mode, any device can only communicate point-to-point with the host that initiates silence, and silence can only be exited by the host that initiates silence. 5. When the host that initiated the silence sends a broadcast message, it can be considered to have exited the silence, and all devices can send data normally. 6. The sending cycle is 50ms and the message is sent 5 times.

5.2 Dedicated instructions

5.2.1 Battery

SID	DID	CMD	Index	DLC	Data	illustrate
0x14	0x1F	0x0E	ERR	8	**D0: Fault code (0x10-0x2F)** 0x10: Total voltage is too high 0x11: Total voltage is too low 0x12: Current is too large 0x13: Single cell voltage is too high 0x14: Single cell voltage is too low 0x15: Battery temperature is too high 0x16: Battery temperature is too low 0x17: Charging voltage is too high 0x18: Charging current is too high	To be compatible with the protocol, you can select D0 to send the fault in a single byte, or select D4-D7 to send the fault code bit by bit, or send them simultaneously.

SID	DID	CMD	Index	DLC	Data	illustrate
					0x19: Charging temperature is too high 0x1A: Charging temperature is too low 0x1B: The battery pack voltage difference is too large 0x1C: Battery temperature difference is too large 0x1D: Battery capacity is too low 0x1E: Short circuit protection 0x1F: Voltage drop protection 0x20: MOS high temperature 0x21: Discharge MOS failure 0x22: Charging MOS failure 0x23: NTC abnormality 0x24: Collection line abnormality 0x25: Low voltage, charging prohibited 0x26: Vehicle code does not match 0x27: The battery is flooded **D1-D2: Reserved** **D3:** bit0: vehicle code does not match bit1: Battery is flooded **D4-D7: Indicated by bit**, 0 means no fault, 1 The sending cycle is 1S, no broadcasting if there is no fault **D4:** bit0: Single cell overvoltage bit1: single cell undervoltage bit2: Battery pack overvoltage bit3: Battery pack undervoltage bit4: Battery pack voltage difference is too large **D5:** bit0: Thermal failure control alarm bit1: Thermal runaway a) and c) triggers bit2: Thermal runaway b) and c) triggers bit3: short circuit protection bit4: discharge overcurrent level 1 bit5: Discharge overcurrent level 2 bit6: Charging overcurrent level 1	

SID	DID	CMD	Index	DLC	Data	illustrate
					bit7: Charge overcurrent level 2 **D6:** bit0: Charging high temperature bit1: low temperature charging bit2: discharge high temperature bit3: Discharge low temperature bit4: MOS high temperature bit5: Battery temperature difference is too large bit6: Battery capacity is too low **D7:** bit0: discharge MOS bit1: charging MOS bit2: voltage difference protection bit3: Acquisition line abnormality bit4: low voltage, charging prohibited bit5: NTC abnormality	
0x14	0x1F	0x10	0x01	7	D0D1: Total capacity 10 mAh D2D3: Remaining capacity 10 mAh D4: Relative capacity percentage, 0-100 D5: Absolute capacity percentage, 0-100 D6: Current allowable continuous discharge current 1 A	Sending cycle 500 ms.
0x14	0x1F	0x10	0x02	8	D0: 0x00 discharge, 0x01 charge, 0x02 standstill D1D2: Current current 0.01 A (the current is always positive, and the charging and discharging can be judged by the above status) D3D4D5: Current total voltage 1m V D6: Maximum battery temperature (positive offset 40 degrees, unit: degrees), actual temperature = D6-40 (°C) D7: minimum battery temperature (positive offset 40 degrees, unit: degrees), actual temperature = D7-40 (°C)	Sending cycle 100ms.

SID	DID	CMD	Index	DLC	Data	illustrate
0x14	0x1F	0x10	0x40	6	D0: Discharge line connected 0- not connected, 1 - connected D1: Address line access 0- not accessed, 1- accessed D2: Charger connected 0- not connected, 1- connected D3: Load connection 0- not connected, 1- connected D4: Discharge MOS switch state 0- closed, 1- opened D5: Charging MOS switch status 0- closed, 1- opened D6: Key line access 0- not connected, 1- connected	Sending cycle 1S.
0x14	0x1F	0x11	0x01	5	D0: 0x00 invalid 0x01 switching 0x02 switching successful 0x03 switching failed D1: 0x00 discharge, 0x01 charge, 0x02 standstill D2: 0x00 allows discharge, 0x01 prohibits discharge D3: 0x00 invalid 0x01: Battery 1 is in a discharged state 0x02: Battery 2 is in a discharged state D4: 0x00 is invalid 0x01: Battery 1 is in a charging state 0x02: Battery 2 is in a charging state	Sending cycle 100ms (CMD 0X42 linkage data).
0x14	0x1F	0x12	0x01	8	D0: Number of strings D1: Number of parallels D2D3: Design capacity 10mAh D4: Maximum allowable charging current 0.4 A D5: maximum allowable continuous discharge current 1 A D6D7: Design cycle number	Sending cycle 500 ms.
0x00-0x1E	0x14	0x14	0x01	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x14	0x01	6	D0D1: Total capacity 10 mAh D2D3: Remaining capacity 10 mAh D4: Relative capacity percentage, 0-100 D5: Absolute capacity percentage, 0-100	Response battery data.
0x00-0x1E	0x14	0x14	0x02	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x14	0x02	8	D0: 0x00 discharge, 0x01 charge, 0x02 standstill	Response battery data.

SID	DID	CMD	Index	DLC	Data	illustrate
					D1D2: Current current 0.01 A (the current is always positive, and the charging and discharging can be judged by the above status) D3D4D5: Current total voltage 1m V D6: Maximum battery temperature (positive offset 40 degrees, unit: degrees), actual temperature = D6-40 (°C) D7: minimum battery temperature (positive offset 40 degrees, unit: degrees), actual temperature = D7-40 (°C)	
0x00-0x1E	0x14	0x14	0x03	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x14	0x03	8	D0: Number of temperature points D1: Battery temperature 1 (positive offset 40 degrees, unit: degrees), actual temperature = D1-40 (°C) D2: Battery temperature 2 (positive offset 40 degrees, unit: degrees), actual temperature = D2-40 (°C) D3: Battery temperature 3 (positive offset 40 degrees, unit: degrees), actual temperature = D3-40 (°C) D4: Battery temperature 4 (positive offset 40 degrees, unit: degrees), actual temperature = D4-40 (°C) D5: MOS temperature (positive offset 40 degrees, unit: degrees), actual temperature = D5-40 (°C) D6: Connector temperature (positive offset 40 degrees, unit: degrees), actual temperature = D6-40 (°C)	Response battery data.
0x00-0x1E	0x14	0x14	0x04	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x14	0x04	8	D0-D3: Accumulated charging milliampere-hours 1mHA D4-D7: cumulative Discharge milliampere-hour 1mHA	Response battery data.
0x00-0x1E	0x14	0x14	0x05	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x14	0x05	1	D0: Electric door lock status 0x00 allows discharge, 0x01 prohibits discharge	Response battery data.
0x00-0x1E	0x14	0x14	0x06	1	D0: Electric door lock status 0x00 allows discharge, 0x01 prohibits discharge	Write battery data, response using general.
0x00-0x1E	0x14	0x14	0x10	4	D0-D3: "READ"	Read battery data.

SID	DID	CMD	Index	DLC	Data	illustrate
0x14	0x00-0x1E	0x14	0x10	6	D0: Discharge line connected 0- not connected, 1- connected D1: Address line access 0- not accessed, 1- accessed D2: Charger connected 0- not connected, 1- connected D3: Load connection 0- not connected, 1- connected D4: Discharge MOS switch state 0- closed, 1- opened D5: Charging MOS switch status 0- off, 1- on D6: Key line access 0- not connected, 1- connected	Response battery data.
0x00-0x1E	0x14	0x40	0x01	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x40	0x01	8	D0: Number of strings D1: Number of parallels D2D3: Design capacity 10mAh D4: Maximum allowable charging current 0.4 A D5: maximum allowable continuous discharge current 1 A D6D7: Design cycle number	Response battery data.
0x00-0x1E	0x14	0x40	0x02	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x40	0x02	8	D0D1: Number of cycles D2D3: Maximum uncharged time (hours) D4D5: Last time not charged (hours) D6D7: Estimated remaining charging time (minutes)	Response battery data.
0x00-0x1E	0x14	0x41	0x03	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x03	8	D0D1: Cell 1 D2D3: Cell 2 D4D5: Cell 3 D6D7: Cell 4	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.
0x00-0x1E	0x14	0x41	0x04	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x04	8	D0D1: Cell 5 D2D3: Cell 6 D4D5: Cell 7 D6D7: Cell 8	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.
0x00-0x1E	0x14	0x41	0x05	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x05	8	D0D1: Cell 9	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.

SID	DID	CMD	Index	DLC	Data	illustrate
					D2D3: Cell 10 D4D5: Cell 11 D6D7: Cell 12	
0x00-0x1E	0x14	0x41	0x06	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x06	8	D0D1: Cell 13 D2D3: Cell 14 D4D5: Cell 15 D6D7: Cell 16	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.
0x00-0x1E	0x14	0x41	0x07	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x07	8	D0D1: Cell 17 D2D3: Cell 18 D4D5: Cell 19 D6D7: Cell 20	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.
0x00-0x1E	0x14	0x41	0x08	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x41	0x08	8	D0D1: Cell 21 D2D3: Cell 22 D4D5: Cell 23 D6D7: Cell 24	Response battery data. The cell voltage refers to the voltage of the battery cells connected in parallel, measured in mV.
0x00-0x1E	0x14	0x42	0x10	4	D0-D3: "READ"	Read battery balancing data.
0x14	0x00-0x1E	0x42	0x10	8	D0: Balanced state of cell 1 D1: Balanced state of cell 2 D2: Balanced state of cell 3 D3: Balanced state of cell 4 D4: Balanced state of cell 5 D5: Balanced state of cell 6 D6: Balanced state of cell 7 D7: Balanced state of cell 8	Response battery data. Balance state 0-Balance off 1-Balance on.
0x00-0x1E	0x14	0x42	0x11	4	D0-D3: "READ"	Read battery balancing data.
0x14	0x00-0x1E	0x42	0x11	8	D0: Balanced state of cell 9 D1: Balanced state of cell 10 D2: Balanced state of cell 11 D3: Balanced state of cell 12 D4: Balanced state of cell 13 D5: Balanced state of cell 14 D6: Balanced state of cell 15 D7: Balanced state of cell 16	Response battery data. Balance state 0-Balance off 1-Balance on.
0x00-0x1E	0x14	0x42	0x12	4	D0-D3: "READ"	Read battery balancing data.

SID	DID	CMD	Index	DLC	Data	illustrate
0x14	0x00-0x1E	0x42	0x12	8	D0: Balanced state of cell 17 D1: Balanced state of cell 18 D2: Balanced state of cell 19 D3: Cell 20 balanced state D4: Balanced state of cell 21 D5: Balanced state of cell 22 D6: Balanced state of cell 23 D7: Balanced state of cell 24	Response battery data. Balance state 0-Balance off 1-Balance on.
0x00-0x1E	0x14	0x43	0x01	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x43	0x01	4	D0D1D2D3: protocol version	Response battery data.
0x00-0x1E	0x14	0x43	0x03	1	D0: Switch power supply battery 0x00 - invalid 0x01 - battery 1 power supply 0x02 - battery 2 power supply	The controller sends control data to the battery. This data can only be sent to battery 1.
0x00-0x1E	0x14	0x43	0x04	4	D0-D3: "READ"	Read battery data.
0x14	0x00-0x1E	0x43	0x04	3	D0D1: Battery idle time (hours) D2: idle capacity percentage, ranging from 0 to 100	Response battery data.
0x00-0x1E	0x14	0x43	0xF2	2	D0: Restart BMS 1- Restart D1: Control BMS sleep 1- effective	Write the battery data using the universal response. Write 0xFF when there is no need to configure individual data.

When Cin and Din are short connected to Pt, and the charger does not communicate, and the battery can output, the output current is limited to 2A, more than 2A, BMS shut off the discharge, and record the historical data, if the battery cannot discharge, then it should be protected normally (not limited by 2A discharge).

When the controller does not read the current limit value, the current limit value of the controller shall prevail; When the controller reads the battery's current limit value, the battery's current limit value shall prevail;

The conditions for determining thermal runaway are as follows: a) The trigger object generates a voltage drop, and the drop value exceeds 25% of the initial voltage; b) The temperature at the monitoring point reaches the maximum operating temperature specified by the manufacturer; c) The rate dT/dt at the monitoring point is $\geq 1°C/s$ and lasts for more than 3 seconds. When a) and c) or b) and c) occur, it is determined that the trigger object has experienced thermal runaway.

6 Appendix

6.1 Random number generation

An independent single-chip microcomputer system cannot achieve random effects without the introduction of a random signal. The random signal here can be user interaction, time, collected external data, etc. We often use pseudo-random number series + random number seeds to achieve random effects, and these random signals are used as random number seeds.

Several common random signals used by microcontrollers to generate random numbers are as follows:

1. Use one AD to collect power supply noise and take the ADC value as the random signal;
2. Add a variable, put it in the main loop, and use this variable as a random signal;

3. Use the clock chip or RTC to obtain the current time and then process it as a random signal;
4. Build a hardware circuit to generate random signals.

Random number generation function:

```
srand(value);
random = rand();
```

6.2 General response RES

Code	illustrate
0x00	Normal, successful
0x01	Instruction Error
0x02	Data length error
0x03	CRC check error
0x20	Not powered on
0x21	Not registered
0x22	
0x23	Not ready
0x24	Not started
0x40	The device does not support firmware update
0x41	The device is busy and cannot update the firmware temporarily.
0x42	The new firmware file does not match the hardware
0x43	The new firmware file does not match the old firmware
0x44	The new firmware version is lower than or equal to the old firmware version
0xFF	Other Errors

6.3 Fault level ERR

Code	illustrate	action
0x00	No trouble	
0x05	Faulty	

6.4 Fault Codes

Code	illustrate	Failure level	Remark
0x00	No trouble		
0x0A	APP is empty and in boot	0x05	The sending cycle is 100ms until the firmware upgrade command is received.

6.5 CRC16 algorithm

CRC16_MODBUS C language implementation is as follows:

```
unsigned short CRC16_MODBUS (unsigned char *pucBuf, unsigned int uwLength)
{
    uint16_t uiCRCValue = 0xffff;
    uint8_t ucLoop;
    uint8_t * pu8Buf = (uint8_t *)pucBuf;
    while (uwLength--)
    {
        uiCRCValue ^= *pu8Buf++;
        for (ucLoop=0; ucLoop<8; ucLoop++)
        {
            if (uiCRCValue & 0x0001)
```

```
            {
                uiCRCValue >>= 1;
                uiCRCValue ^= 0xA001;
            }
            else
                uiCRCValue >>= 1;
        }
    }
    uiCRCValue ^= 0x0000;
    return uiCRCValue;
}
```

6.6 Segment code display table

character	show	character	show	character	show	character	show
0		A		K	\	U	
1		B		L		V	\
2		C		M	\	W	\
3		D		N		X	\
4		E		About		Y	
5		F		P		Z	\
6		G	\	Q		α	\
7		H		R		β	\
8		I	\	S	\	Γ	
9		J		T		-	

6.7 Firmware Upgrade Symbol Table

symbol	Numeric	illustrate
SOH	0x01	128 Byte Packet
STX	0x02	1024 Byte Packet
EOT	0x04	End of transfer
ACK	0x06	Response

symbol	**Numeric**	**illustrate**
NAK	0x15	Retransmission required
CA	0x18	Cancel Upgrade
C	0x43	Request Packet